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INCA-LLM: Innovative Network Slicing in 5G/6G Core Auto-Configuration with LLM

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Abstract
This work presents an initial approach to automation using Large Language Models, with the main objective of automatically configuring the 5G core for proper network slicing deployment. First, it explains the key concepts needed to understand 5G-LLM integration. Subsequently, it presents our proposal, called INCA-LLM, which uses In-Context Learning to automatically generate the configuration files for the 5G core. INCA-LLM applies few-shot prompting strategy for the first proof-of-concept tests. Preliminary results show that the generated files are suitable for deployment, with a 79.54% similarity to the expert-created configuration.

***Keywords*—Large Language Models, In-Context Learning, Few-shot prompting, Auto-Configuration, Next-Generation Networks, 5G/6G Core, Network Slicing.**

I. INTRODUCTION

Large Language Models (LLMs) have demonstrated strong capabilities in domains such as finance, law, and medicine, yet their adoption in highly technical areas like telecommunications remains limited. The configuration of Next-Generation Networks (NGNs), particularly in 5G and beyond, demands precise domain expertise and involves processes that are often manual, time-consuming, and prone to error. Even small mistakes can lead to service disruptions and significant costs, while the scarcity of qualified experts further increases the burden of network management. These challenges highlight the urgency of developing automation mechanisms capable of reducing human dependency, improving accuracy, and ensuring operational efficiency. Properly adapted, LLMs can help address these limitations by supporting configuration-intensive tasks with greater reliability.

At the same time, the complexity introduced by emerging paradigms such as Software-Defined Networking (SDN), Network Function Virtualization (NFV), Multi-access Edge Computing (MEC), and Network Slicing (NS) has rendered traditional management approaches insufficient. In this context, LLMs are positioned as enablers of self-managing networks, contributing to faster deployment, lower operational costs, and reduced human error, in line with the vision of Zero-touch network and Service Management (ZSM). Building on this perspective, this work introduces Innovative Network Slicing in 5G/6G Core Auto-Configuration with LLM (INCA-LLM), a framework for managing network slicing in industrial contexts. The system automates the generation of 5G core network configurations using data from the Amarisoft core, translating high-level user inputs into precise outputs while mitigating risks such as hallucinations and misconfigurations. By addressing one of the most resource-intensive aspects of network management, INCA-LLM contributes to more reliable, scalable, and efficient next-generation networks.

This paper is structured as follows: Section II reviews related work; Section III outlines key concepts for 5G-LLM integration; Section IV details the INCA-LLM framework; Section V presents validation and results; and Section VI concludes with insights and future directions.

II. RELATED WORKS

Large Language Models are playing an increasingly relevant role in automating NGNs. Their effectiveness depends on high-quality, domain-specific datasets, such as those based on 3rd Generation Partnership Project (3GPP) Technical Specifications (TS) [4]. Tools like 5G Instruct Forge [1] help to generate such datasets, enabling fine-tuning of models like LLaMA and Mistral for 5G tasks.

Managing NGNs, especially for network slicing, is complex due to the wide range of functionalities and the shortage of experts [5] [6]. This has led to the rise of Intent-Based Networking (IBN), where user-defined goals guide automated network management. LLMs are being explored to interpret these intentions and generate configurations, as seen in works like SliceGPT [7] and

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LLM-xAPP [8], which apply LLMs to slice orchestration and radio resource management.

Our work builds on these ideas by focusing on automating the generation of 5G Core Network (5G-CN) configurations, specifically for the Amarisoft core. This is a key component for enabling IBN and ZSM. We emphasize accurate data collection to ensure reliable configuration generation, as errors could affect critical services. The work also presents integration methods with the 5G network and introduces our approach using In-Context Learning (ICL).

III. SYSTEM IMPLEMENTATION

A. 5G-LLM Integration Keys

The integration of LLM in a 5G network requires experience in both areas. Therefore, two key factors are broken down below: (1) LLM inference and utilization, and (2) 5G system and LLM integration.

LLM inference and utilization

Focusing on inference and utilization, note that LLMs specialize in the well-known prompt engineering. Prompt engineering is the process to generate desired outputs, specifically prompt creation requires no additional training and produces instant results based on user input. This represents a straightforward approach to using LLMs. Prompt engineering features several techniques, but in this work we focus on In-Context Learning (ICL). ICL uses formatted natural language prompts and integrates task descriptions and examples to guide LLMs in executing tasks. This approach allows LLMs to recognize and perform new tasks using contextual information. In this regard, four strategies are shown as follows:

1) **Zero-shot prompting:** Provide a clear instruction describing the task, e.g. "Generate a configuration file based on the following requirements...".
2) **Few-shot prompting:** Include 2-3 examples from the several triads directly in the prompt to show the LLM the desired input (natural language description) and output (configuration file) format. This approach is crucial for the task as it allows the model to learn the desired pattern.
3) **Role-playing:** Assign a specific role to the model, e.g. "You are an expert engineer in 5G networks specializing in core configurations. Your task is to generate a configuration file..."
4) **Output format instructions:** Explicitly request that the output conform to the configuration file structure. If possible, provide a template or schema description within the prompt to guide the model.

5G system and LLM integration

From a theoretical-practical framework, the standard 5G Service Based Architecture (SBA) is made up of two parts (RAN and CN). Additionally, there are multiple parts that can be considered within the architecture, such as Near-Real-Time RAN Intelligent Controller (Near-RT RIC) and IP Multimedia Subsystem (IMS) Core. From this point on, all of them will be considered as a 5G system. Fig. 1 shows the 5G System and also the relevant interfaces (green arrows) and IMS connections (purple elements).

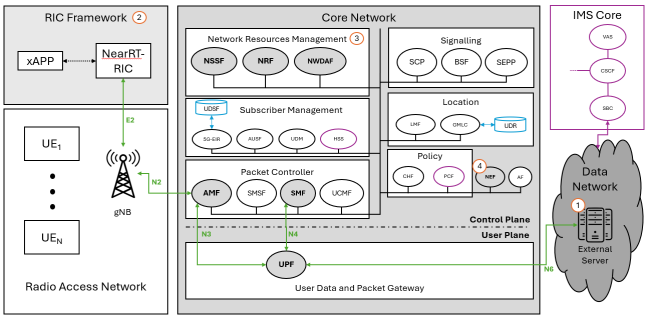

Figure 1. 5G-LLM Integration on 5G System with NearRT-RIC, 5G Core Service Based Architecture and IMS Core

In addition, 5G network functions related to network slicing management need to be well understood (highlighted and in bold). These are defined below:

- **Network Slice Selection Function (NSSF):** It selects the appropriate slice network for a User Equipment (UE) based on service requirements and policies.
- **Network Repository Function (NRF):** It maintains a repository of available network functions and their capabilities, allowing other network functions to discover and communicate with the required functions. It is essential for the management and discovery of network functions that can be used in different network slices.
- **Network Data Analytics Function (NWDAF):** It provides analysis of network data to improve efficiency and performance in real time, e.g. resources for optimisation of resource allocation of different slices.
- **Access and Mobility Management Function (AMF):** It manages the mobility and access of the UE, interacting with the NSSF to ensure that the UE connects to the appropriate network slice according to its service needs.
- **Session Management Function (SMF):** It manages UE data sessions, including session configuration within network slices, ensuring that they meet specific Quality of Service (QoS) requirements.
- **User Plane Function (UPF):** It manages user data traffic for different network slices, including routing, packet forwarding, policy enforcement and QoS management of each slice.
- **Network Exposure Function (NEF):** It exposes network capabilities and services to external and secure applications, enabling the integration of third party services.

We highlight four integration methods (marked numerically on Fig.1):

1) **External server:** It is the most versatile method due to the fact that can encompass both RAN and CN. It consists of deploying the LLM service on a server external to the 5G system. Once requested, it communicates with the 5G system to auto-configure

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or modify parameters to optimize network slices according to the user requirements. These servers can be deployed at the Edge or Cloud.

2) **xApp:** This approach involves creating a framework that integrates an LLM as an xApp running on the NearRT-RIC of O-RAN, which is part of the RAN. Therefore, it allows for the monitoring and adaptive management of radio resources, translating high-level UE intentions into RAN configurations, such as optimizing PRBs based on UE needs. It is important to note that this method focuses solely on the RAN.
3) **Network Resource Management:** This method, focused solely on the CN, involves the implementation or enhancement within NWDAF. Essentially, the LLM receives data collected from other NFs, analyzes it, and optimizes network and resource management internally within the CN.
4) **Network Exposure:** It is similar to the external server. The main difference is that the requirements or prompt must be provided to the system via NEF.

B. INCA-LLM

Testbed

INCA-LLM is deployed on an edge server as shown in Fig. 2, as this setup offers an efficient and practical way to generate initial configurations. It also allows for future extensions to both RAN and CN. INCA-LLM focuses on Amarisoft core due to its compact configuration format, which uses fewer files compared to alternatives like open5GS or OAI. As shown in Fig. 2, the system integrates INCA-LLM with the 5G network and demonstrates a configuration with multiple slices, being the main objective of the auto-generated configuration files. In this case, the network would support three slices. The first slice is for typical internet traffic and data calls over the eMBB slice with a low priority 5QI (e.g., 5QI 9). The second slice, called URLLC, is for critical industrial traffic and will use a high priority with 5QI 90. Finally, the third slice, called mMTC, will contain the monitoring traffic, generated by multiple industrial devices, with a medium priority and 5QI 86.

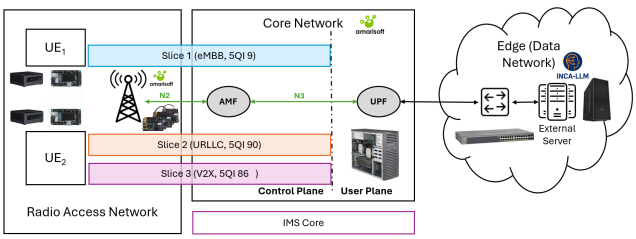

Figure 2. Integration 5G System with INCA-LLM

Fig. 2 also illustrates the components of the testbed where the system is deployed and evaluated. The system has been deployed on a testbed to evaluate its functionality and results. Primarily, the INCA-LLM service is deployed on an external server, specifically a DELL 1445 server. The 5G-Amarisoft system is implemented on a general-purpose computer equipped with two PCIe SDR50. Finally, to verify connectivity, we use two UEs consisting of an Intel NUC BXNUC10I7FNH2 paired with a Quectel RM500Q-GL board.

INCA-LLM Implementation

INCA-LLM is designed and implemented following the principles of ICL. It implements two modes. Both are based on the Few-shot prompting strategy, using the default configuration of Amarisoft and a multiple slicing configuration files that contains the necessary settings for the correct deployment of the slices. Our implementation, see Fig. 3, is based on the LLM Ollama and the model deepseek-r1:14b have been tested. These mode are:

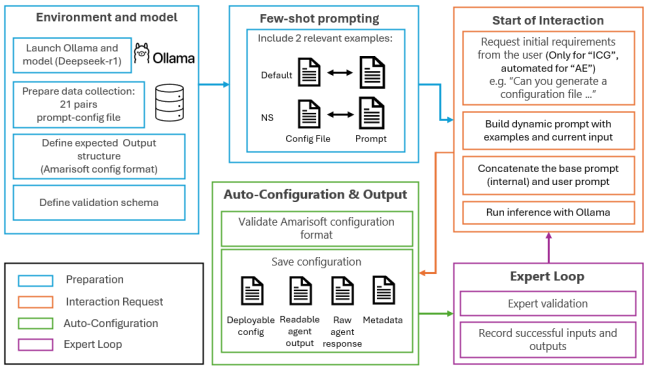

Figure 3. INCA-LLM implementation: Network slicing auto-configuration for 5G/6G CN.

1) **Interactive Configuration Generation (ICG):** This mode allows for the structured generation of CN configurations, using the two cited configurations as references while maintaining the Amarisoft core configuration format (following [9])). The Amarisoft format has a high level of complexity due to its key-value structure (similar to JSON) and C directives. Through user-model interaction, the LLM service generates a configuration based on the requirements provided in natural language.
2) **Automated Evaluation of Prompts (AEoP):** This mode consists of a systematic process designed to evaluate multiple iterations based on different prompt designs on the model, comparing their outputs with the expected configurations previously defined by an expert in 5G core configuration of Amarisoft. It works from a set of test cases, where each prompt is evaluated in parallel, and their outputs are automatically validated through structural comparison with reference files.

IV. VALIDATION & RESULTS

The INCA-LLM system is evaluated through four validation steps:

1) **Format validation (V_1):** It checks the configuration file generated by the model complies with the expected format for the Amarisoft core configuration, ensuring that the syntactic structure is valid.
2) **Expert validation (V_2):** It implies that an expert manually reviews the configurations generated by

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the model to ensure that the values, parameters, and structure are appropriate, secure, and consistent, strictly adhering to the user's requirements.

3) **Deployment (V_3):** involves deploying the configuration in the testbed and observing if the 5G system exhibits any deployment failures. This validation ensures that the configuration is not only theoretically correct but also operationally functional.
4) **Structural similarity (V_4):** This metric quantitatively evaluates the degree of alignment between the model-generated configuration and the reference configuration provided by experts. It considers the hierarchical organization, presence of mandatory fields, and consistency of internal object relationships. Formally expressed as Eq. 1.

$$S(A,B) = \frac{2M}{|A|+|B|} * 100 \tag{1}$$

where A and B represent the normalized reference and generated configurations, M denotes the number of matching characters identified in the longest common subsequences, and $|A|$, $|B|$ correspond to their respective lengths. The resulting value $S(A,B) \in [0,100]\%$ provides a robust indicator of structural similarity, with values closer to 100 % denoting stronger alignment between the two configurations.

The results obtained are condensed in Table I, showing the validations passed and the percentage of similarity. It should be noted that the percentage of similarity is never 100% because it is a file generated by the model and the inference obtained, so there may be some differences from the reference configurations, such as simple comments within the configuration.

Table I
PRELIMINARY RESULTS OF INCA-LLM SYSTEM.

Test	V_1	V_2	V_3	V_4
T_1	✓	✓	✗	74.51%
T_2	✓	✓	✓	80.23%
T_3	✓	✓	✓	73.75%
T_4	✓	✓	✗	87.86%
T_5	✓	✓	✗	71.08%
T_6	✓	✓	✓	89.82%

The results show that 100% of the generated configurations correctly follow the Amarisoft structure. At the same time, the expert states that the parameters referred to in the prompts are correctly modified. But as we can see, there are still some parameters not specified that are configured wrongly, making 50% of the files generated for testing invalid for deployment.

V. CONCLUSION

In this paper, we introduce the integration of a 5G/6G system with an LLM service, called INCA-LLM. The proposed integration consists of deploying the service at the edge with the aim of automatically generating configuration files for the network core, focusing mainly on network slice deployments. INCA-LLM is based on ICL, specifically on the few-shot prompting strategy. Our deployment and evaluations show that INCA-LLM, without the need for training (fine-tuning) and specialized in Amarisoft, correctly generates the files according to user specifications, demonstrating significant structural similarities with the files provided by the 5G network configuration expert (approximately 79.54%). This work advances the use of LLMs in telecommunications and provides a system that can be extended to other software such as OAI or open5GS. Looking ahead, our research has important implications for the development of future networks like 6G, which will rely on technologies such as zero-touch management systems.

The research lines to be followed involve the extension of LLMs for the generation of Amarisoft files corresponding to the rest of the 5G system (gNB, DRB data base, IMS Core and UE data base). Subsequently, the addition of auto-configuration without the need for expert review will be implemented, and finally, monitoring and new LLM agents will be deployed as specified in the integration methods to optimize the network.

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